Open problems:
Multiple equivalent characterizations of the Schrödinger equation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Injectivity suffices in finite dimension.** For $\dim \mathcal{H} < \infty$, an affine map $T : D(\mathcal{H}) \to D(\mathcal{H})$ injective on the pure states $P(\mathcal{H})$ is bijective on them. The brief's reason, "the compactness of $P(\mathcal{H})$ will force T to be bijective", does not suffice; the log points to invariance of domain on the manifold $P(\mathbb{C}^n)$. Not formalized (only the version with surjectivity is). (Brief, Remark after Proposition 1; log, "Problems found in the brief", finite-dimensional part.)

2. **Time-dependent evolution.** The equivalence of DR-SCEQ, DR-UNIT, DR-EV, DR-PROB and DR-INFO without EVTI, for two-parameter families $\Phi_{t_0 \to t_1}$ and a time-dependent Hamiltonian $H(t)$, as DR-SCEQ is written. The brief only says "We are going to restrict the dynamics to the time-independent case"; the formalization uses a constant H. (Brief, Sec. 3, before Condition EVTI, and Condition DR-SCEQ; log, "Differences from the brief".)

3. **Isolation as determinism and reversibility.** "Conceptually, all these are equivalent to saying that the system is isolated, as future and past states are determined, and only determined, by the present state of the system." Isolation is not defined, so this equivalence is not stated as a theorem. (Brief, Sec. 6.)

From the formalization

Beyond finite dimension the formalization has a part in any dimension, which covers unbounded H; it leaves open the item "Shift semigroup counterexample" below.

1. **Shift semigroup counterexample.** The right shifts V_t on $L^2(0, \infty)$ give an evolution $\Phi_t(\rho) = V_t \rho V_t^\dagger$ that maps pure states to pure states and preserves the Born rule and the entropy, but satisfies neither DR-EV nor DR-UNIT. It shows that the reconstructed past clauses of DR-PROB and DR-INFO are needed in infinite dimension. Formalized only as a single map (`shift_preserves_not_unitary`), not as a semigroup satisfying `IsEvolution`. (Log, "Hilbert spaces of any dimension", "Problems found in the brief".)

2. **Proposition 1 for equal mixtures.** Proposition 1 is proved for maps affine under all convex combinations. A version assuming only $T(\frac{1}{2}\rho + \frac{1}{2}\sigma) = \frac{1}{2}T\rho + \frac{1}{2}T\sigma$ "would need continuity first, to pass from dyadic weights to all weights"; not formalized. (Log, "Hilbert spaces of any dimension", "Differences from the brief".)